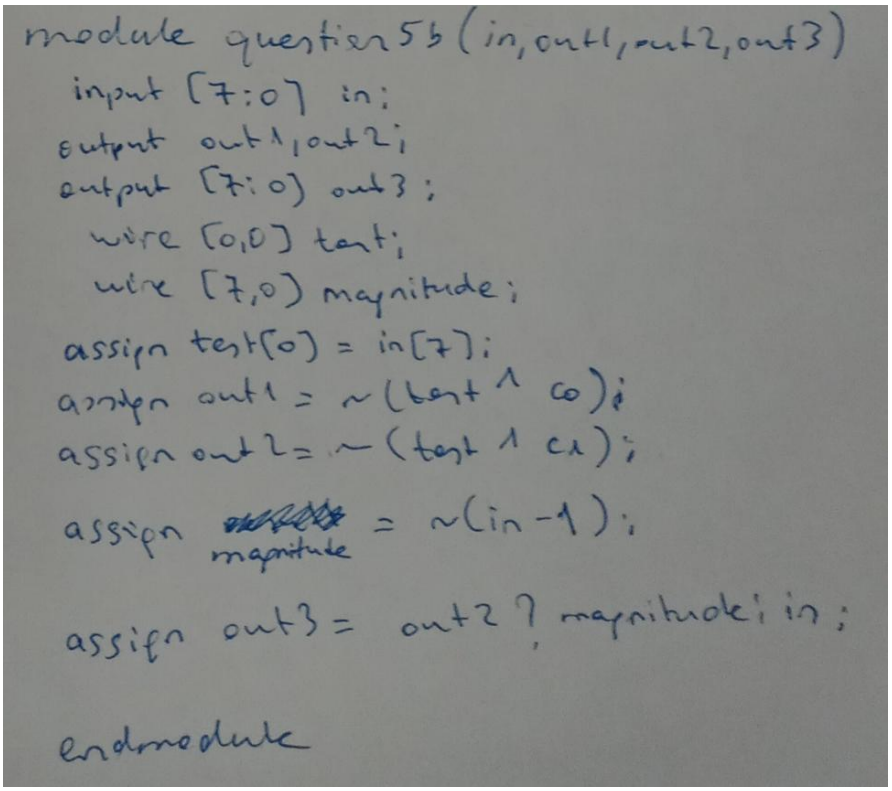
LAB4 DIG. DES. 19th decembre

Here's a previous exam question:

Q5b) Design a magnitude displayer with the following specifications. It will be single input single output circuit. Input to the system is an eight-bit number which can be positive or negative in two's complement form. Output will display the magnitude of the input number. Hint: If the input number is negative, you can find its magnitude by converting it back from two's complement form.

A) **(10 points)** Implement the designed combinational circuit in Verilog.

Solution of A is below:



```
module question5b (in, out1, out2, out3)
  input [7:0] in;
output out1, out2;
output [7:0] out3;
  wire [0,0] test;
  wire [7,0) magnitude;
assign test[0] = in[7];
assign out1 = ~(test ^ c0);
assign out2 = ~(test ^ c1);
assign [illegible] magnitude = ~(in-1);
assign out3 = out2 ? magnitude: in;
endmodule
```

LAB TASKS:

B) Write a testbench file for q5.
C) Show example outputs using repeat.
D) Find limits of 8-bit number representation with two's complement.

HINT:

An example output:

Above line shows exact number.

Bottom line shows magnitude.

Middle lines Show if a number is positive or negative.

